SAI SIVA NAGA DHEERAJ GURAZADA

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SUMMARY

Computer Engineering MS student at NC State with **2.5+ years** of hardware/firmware validation on safety-critical **Airbag ECUs** at **Bosch** (HIL/SIL, CAN/UDS, JTAG, Python/Perl automation), plus **Verilog RTL design**, **SystemVerilog/UVM** verification, FPGA prototyping, and an **out-of-order CPU simulator** in C++. Seeking **Summer 2027** internships in **Design Verification, Silicon and System Validation, or Embedded Firmware Validation**.

EDUCATION

North Carolina State University **Aug 2026 – May 2028 (Expected)**
Master of Science in Computer Engineering *Raleigh, NC*
- **Coursework:** ASIC and FPGA Design with Verilog, Microprocessor Architecture, Parallel Computer Architecture.

SASTRA Deemed University **Aug 2019 – May 2023**
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 8.56/10 *Thanjavur, India*

TECHNICAL SKILLS

- **Design & Verification:** Verilog, SystemVerilog, UVM, SVA, Functional Coverage, Constrained-Random Verification, Scoreboards, Reference Models, AMBA APB, FSM, FIFO, Synthesis, FPGA Prototyping.
- **Validation & Lab Debug:** HIL, SIL, Requirements-Based Testing, Test Automation, Fault Injection, Root-Cause Analysis, JTAG, Oscilloscope, Logic Analyzer.
- **Embedded & Protocols:** ECU Validation, CAN, CAN FD, UDS, J1939, UART, DBC, DTC Diagnostics.
- **Computer Architecture:** Pipelining, Out-of-Order Execution, Register Renaming, ROB, Issue Queue, Caches.
- **Programming & Tools:** C, C++, Python, Perl, MATLAB, Linux, Jenkins (CI/Regression), Questa, ModelSim, Vivado, CANoe, ECU-TEST, TurboLIFT.

EXPERIENCE

Bosch Global Software Technologies **Jan 2024 – Jul 2026**
Associate Software Engineer – Embedded Systems (Verification and Validation) *Coimbatore, India*
- Validated **ISO 26262** safety-critical **Airbag ECU** hardware and firmware against requirements, developing **Python and Perl** test scripts to verify **CAN/UDS** communication across **HIL and SIL** benches.
- Root-caused ECU failures across hardware and firmware by injecting faults, reading **DTCs via UDS**, and probing signals with **oscilloscopes, logic analyzers, and JTAG**; reported **50+ defects**.
- Automated **nightly regression** in **Jenkins** across HIL and SIL benches, cutting manual verification effort by **30%** (HIL) and **50%** (SIL).
- Built a **Python DBC analysis tool** and an **AI-assisted code review assistant**, reducing redundant manual work by **40%**.

Maven Silicon **Jun 2023 – Dec 2023**
Design and Verification Trainee *Remote*
- Built a **UVM** testbench for an **AMBA APB slave RAM** with an active agent (sequencer, driver, monitor), scoreboard with reference model, and coverage subscriber; ran directed and **400 constrained-random** transactions to reach **95% functional coverage**.
- Designed a synthesizable **1x3 packet router** in **Verilog** with header-based address decoding, FSM control, per-port **FIFO** buffering, parity checking, and reset handling.

Yuan Ze University **Feb 2023 – Jul 2023**
RTL Design and FPGA Research Intern *Taoyuan, Taiwan*
- Designed an image dehazing hardware accelerator in **Verilog** with a **line-buffer-less architecture**, cutting on-chip memory for airlight estimation from **4–8 line buffers to zero** versus prior designs.
- Implemented and validated synthesizable RTL on a **Xilinx PYNQ-Z2 FPGA**, with an end-to-end **MATLAB-to-RTL-to-Vivado** flow.

PROJECTS

Out-of-Order Superscalar Processor Simulator | C++ **Aug 2026 – Present**
- Building a cycle-accurate **out-of-order superscalar** simulator with **register renaming**, **issue queue**, and **ROB**; trace-driven, sweeping width, ROB, and issue-queue sizes to measure **IPC**.

PUBLICATIONS

Revisiting the Utility of Spectral Measures on Spoken Text Recognition **IEEE ICASSP, May 2023**